

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q13-0222

White Swan 138 kV
March 31, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0222. The Interconnection Customer proposes a 66.7 MW solar photovoltaic facility interconnecting at the White Swan 138 kV substation in the Pennsylvania area. The request was received on March 31, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 136 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$50,190,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The CLEAR CROSSING - PULPGEN 138 kV Line is overloaded to 138.3% of its 190 MVA emergency rating for the loss of the CLEAR SPRINGS 500/345 kV TX. The project contributes 6.7% DFAX (4.5 MW) to this overload.

CLEAR CROSSING - WINDSTONE 138 kV: 131.9% loading (206 MVA) following the outage of the CLEAR SPRINGS 500 kV/345 kV Transformer; DFAX = 6.7%.

For the loss of W ALLATIN - WALNT 345 kV Line, the VERDE MAR 2605 - CROTON FALLS SUBSTATION 138 kV Line loads to 130.7% of its 295 MVA rating. Project DFAX: 8.5%.

The POLK ST - RAILHEAD 138 kV is overloaded to 129.0% of its 275 MVA emergency rating for the loss of the WALNUT - WEST GALLATIN 345 kV. The project contributes 11.1% DFAX (7.4 MW) to this overload.

For the loss of CLEAR SPRINGS 500 kV/345 kV Transformer, the WINDSTONE - WALDEN NORTH GENERATION STA 138 kV loads to 127.9% of its 225 MVA rating. Project DFAX: 6.7%.

For the loss of CLEAR SPRINGS SUBSTATION 500/345 kV Autotransformer, the LOS ANGELES HTS - VAN WAGNER 138 kV loads to 127.3% of its 244 MVA rating. Project DFAX: 6.7%.

For the loss of CLEAR SPRINGS 500/345kV Auto, the LOS ANGELES HTS - WALDEN N GENERATION STATION 138 kV loads to 126.4% of its 239 MVA rating. Project DFAX: 6.7%.

POLK ST - VAN WAGNER 138 kV: 118.1% loading (289 MVA) following the outage of the W GALLATIN - WALNUT SUBSTATION 345 kV; DFAX = 11.1%.

The JUDITH GAP WIND FARM - SPLIT ROCK 138 kV Line is overloaded to 101.5% of its 219 MVA emergency rating for the loss of the WALNUT - W GALLATIN 345 kV. The project contributes 5.7% DFAX (3.8 MW) to this overload.

The VERDE MAR 2605 - JUDITH GAP WIND FARM 138 kV is overloaded to 100.5% of its 228 MVA emergency rating for the loss of the WALNUT - WEST GALLATIN 345 kV. The project contributes 5.7% DFAX (3.8 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CLEAR CROSSING - PULPGEN 138 kV Line	Rebuild line with higher-capacity conductor	\$6,707,000	100% (\$6,707,000)
WINDSTONE - CLEAR CROSSING SUBSTATION 138 kV Line	Rebuild line with higher-capacity conductor	\$2,621,000	100% (\$2,621,000)
CROTON FALLS - VERDE MAR 2605 138 kV	Rebuild line with higher-capacity conductor	\$7,753,000	53% (\$4,097,000)
POLK ST - RAILHEAD 138 kV	Rebuild line with higher-capacity conductor	\$6,006,000	100% (\$6,006,000)
WINDSTONE - WALDEN NORTH GENERATION STA 138 kV	Rebuild line with higher-capacity conductor	\$7,740,000	100% (\$7,740,000)
VAN WAGNER - LOS ANGELES HTS 138 kV	Rebuild line with higher-capacity conductor	\$3,787,000	100% (\$3,787,000)
WALDEN N GENERATION STA - LOS ANGELES HEIGHTS 138 kV	Rebuild line with higher-capacity conductor	\$3,469,000	100% (\$3,469,000)
POLK ST - VAN WAGNER 138 kV	Rebuild line with higher-capacity conductor	\$7,935,000	100% (\$7,935,000)
SPLIT ROCK - JUDITH GAP WIND FARM 138 kV	Reconductor line	\$3,037,000	100% (\$3,037,000)
VERDE MAR 2605 - JUDITH GAP WIND FARM 138 kV	Reconductor line	\$4,791,000	100% (\$4,791,000)
	Total allocated to project		\$50,190,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.